

Salah Eddine Oukil

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SUMMARY

DevOps Engineer and MSc Computer Science student with professional experience in automating cloud infrastructure and high-stakes environment management. Proven expertise in Kubernetes orchestration, Infrastructure as Code (Terraform, Ansible), and AWS. Successfully led disaster recovery missions and conducted security audits for London-based Fintech clients. Currently seeking a DevOps or Platform Engineering role to apply my expertise to a high-growth engineering team.

PROFESSIONAL EXPERIENCE

THE TRYBE AGENCY

DevOps Engineer

Algiers, Algeria

Sep 2024 – Jan 2026

- Developed a custom GitHub Action using the Laravel Forge API to enable declarative, YAML-based infrastructure provisioning; eliminated manual UI bottlenecks and provided version-controlled environments for multiple dev teams.
- Led critical Disaster Recovery for a legacy Kubernetes (kops) cluster on AWS following a total system deletion; reconstructed the entire microservices architecture and manifests without existing documentation.
- Conducted a Security & Infrastructure Audit for a London-based Fintech client; delivered a strategic action plan that hardened AWS cloud infrastructure in accordance with CIS benchmarks, secured CI/CD pipelines, and improved backup protocols.
- Architected an enterprise Monitoring & Observability stack using Prometheus, Grafana, and Loki; implemented custom alerting rules and dashboards to monitor 50+ client servers and applications.
- Automated Infrastructure-as-Code (IaC) workflows using Terraform and Ansible to manage server provisioning, configuration, and security hardening, ensuring reproducible environments across a diverse client portfolio.
- Designed CI/CD pipelines for microservices using GitHub Actions and AWS ECS; implemented robust backup solutions with Point-in-Time Recovery (PITR) to ensure high availability and data integrity.

EFREI

Research Intern

Paris, France

May 2025 – Jul 2025

- Developed an end-to-end multimodal machine learning pipeline integrating high-dimensional gene expression and clinical datasets for cancer diagnosis and prognosis.
- Engineered a deep learning-based autoencoder architecture to perform feature extraction and dimensionality reduction, significantly improving predictive performance and survival prognosis for Multiple Myeloma patients.
- Optimized data processing workflows for large-scale genomic data using PyTorch, Keras, and NumPy, ensuring efficient model training, scalability, and reproducibility of research results.
- Implemented advanced data structures and algorithms to handle complex multimodal inputs, streamlining the transition from raw clinical data to actionable predictive insights.

GEN42

Full Stack Developer

Algiers, Algeria (Remote)

Sep 2023 – Apr 2024

- Containerized the entire application stack using Docker and Docker Compose, ensuring environment parity between local development, staging, and production systems.
- Architected and implemented CI/CD pipelines to automate the testing and deployment of a social media management platform, significantly reducing manual deployment errors.
- Developed scalable backend services using Django to handle complex data retrieval and processing from third-party social media APIs.
- Streamlined the developer onboarding process by creating standardized local development environments, reducing the time required for environment setup by new team members.

EDUCATION

Aston University

MSc in Computer Science

London, UK

Jan 2026 – Present

Higher National School of Computer Science

Master's and State Engineer Degree in Computer Science

Algiers, Algeria

2020 – 2025

PERSONAL PROJECTS

High-Availability Kubernetes Homelab

Proxmox, k3s, etcd, Longhorn, Traefik

- Architected a multi-node Kubernetes (k3s) cluster hosted on a Proxmox VE environment, utilizing a dedicated external etcd node for high-availability control plane state management.
- Implemented advanced networking and ingress using Kube-Vip for control plane load balancing and Traefik as the Ingress Controller to manage traffic for 10+ self-hosted microservices.
- Configured distributed block storage using Longhorn, providing persistent volume claims (PVCs) for stateful applications, backed by a custom Debian-based NAS with ZFS storage pools.
- Deployed a comprehensive media and utility stack (Jellyfin, Arrs) using GitOps-adjacent workflows, ensuring system-wide observability and automated data management.

Redis-Compatible In-Memory Database

C++, TCP/IP, RESP, Non-blocking I/O

- Engineered a high-performance key-value store from scratch in C++, implementing a custom TCP server to handle concurrent client connections.
- Developed a protocol parser for RESP (Redis Serialization Protocol), enabling compatibility with standard Redis clients and ensuring efficient data serialization.
- Implemented an event-driven architecture using an event loop with non-blocking I/O, allowing the server to handle multiple asynchronous requests with minimal latency.
- Designed in-memory data structures optimized for fast retrieval and memory efficiency, gaining deep insights into systems-level resource management and network programming.

SKILLS

Cloud & Infrastructure	AWS (EC2, S3, RDS, ECS, EKS), Digital Ocean, Laravel Forge, Terraform, Ansible.
Containers & Orchestration	Kubernetes (EKS, k3s, kops), Docker, Docker Compose, Helm, Kube-Vip, Traefik, Longhorn.
CI/CD & Automation	GitHub Actions, CircleCI, Jenkins, REST APIs, Bash Scripting.
Monitoring & Logging	Prometheus, Grafana, Loki, ELK Stack (Elasticsearch, Logstash, Kibana).
Programming Languages	Python, Golang, C/C++, Java, Kotlin, JavaScript/TypeScript.
Databases	PostgreSQL, MySQL, Oracle, SQLite, Redis.
Web & Frameworks	Node.js, NestJs, Django, Flask, React, Redux, Nextjs.
Languages	English (Professional Proficiency, IELTS 8.0), Arabic (Native), French (B2).